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more energy than longer wavelength. Electromagnetic spectrum consists of 8 types of radiations such as cosmic rays, gamma rays, X rays, U-V rays, Visible light spectrum, infrared rays, electric rays and radio rays (Figure 13. 5).



Light is extremely variable and if radiation is evenly distributed over the globe it is sufficient to melt 35 m thick ice layer.

Properties of Light

1. Light is a transverse electromagnetic wave.
2. It consists of oscillating electric and magnetic fields that are perpendicular to each other and perpendicular to the direction of propagation of the light.
3. Light moves at a speed of 3×10^8 ms^{-1}
4. Wavelength is the distance between successive crests of the wave.
5. Light as a particle is called **photon**. Each photon contains an amount of energy known as **quantum**.
6. The energy of a photon depends on the frequency of the light (Figure 13. 6).

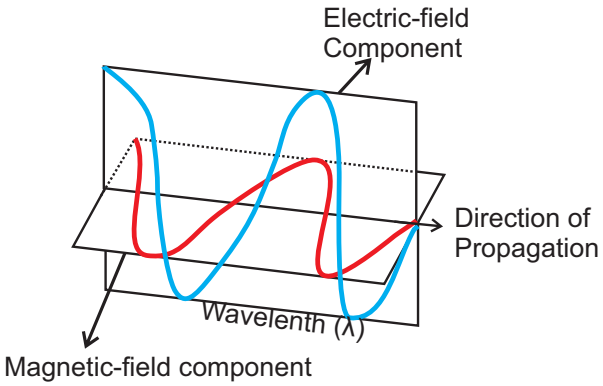


Figure 13.6: Oscillation of electric and magnetic vectors in light

13.5 Photosynthetic Unit (Quantasome)

Quantasomes are the morphological expression of physiological photosynthetic units, located on the inner membrane of thylakoid lamellae. Each quantasome measures about 180 Å × 160 Å and 100 Åthickness. In 1952, **Steinman** observed granular structures in chloroplast lamellae under electron microscope. Later, **Park** and **Biggins** (1964) confirmed these granular structures as physiological units of photosynthesis and coined the term Quantasome. According to them one quantasome contains about 230 chlorophyll molecules. A minimum number of chlorophyll and other accessory pigments act together in a photochemical reaction to release one oxygen or to reduce one molecule of CO_2. It constitutes a photosynthetic unit. (Figure 13.7) **Emerson** and **Arnold** (1932) based on flashing light experiment found 2500 chlorophyll molecules are required to fix one molecule of CO_2. However, the reduction or fixation of one CO_2 requires 10 quanta of light and so each unit would contain 1/10 of 2500 i.e. 250 molecules. Usually 200 to 300 chlorophyll molecules are considered as a physiological unit of photosynthesis. According to Emerson 8 quanta of light are required for the release of one oxygen molecule or reduction of one Carbon dioxide molecule. The quantum yield is 1/8 or 12 %.

13.6 Absorption Spectrum and Action Spectrum

13.6.1 Absorption Spectrum

The term absorption refers to complete retention of light, without reflection or